

Introduction to Glacial Isostatic Adjustment

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Outline

Two-Part Lecture:

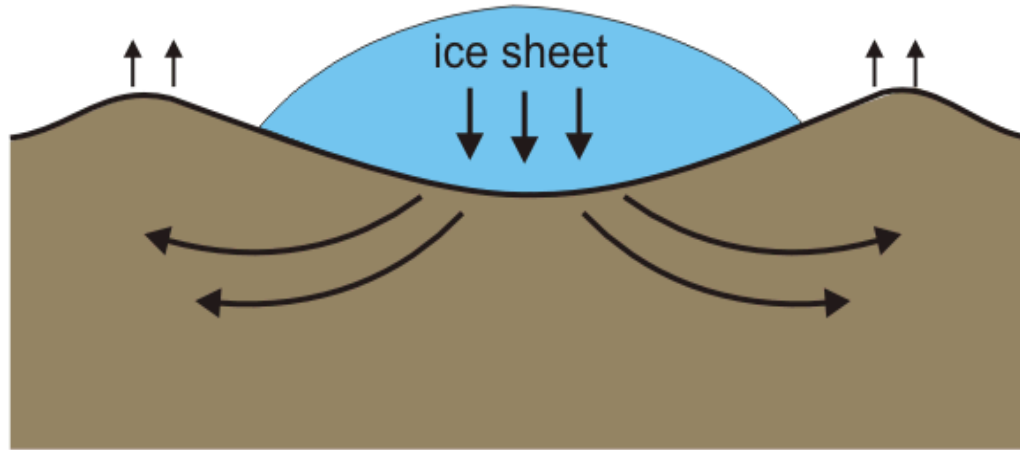
Broad overview of GIA and model components.
(~30 mins)

GIA modelling and model applications (~50 mins).

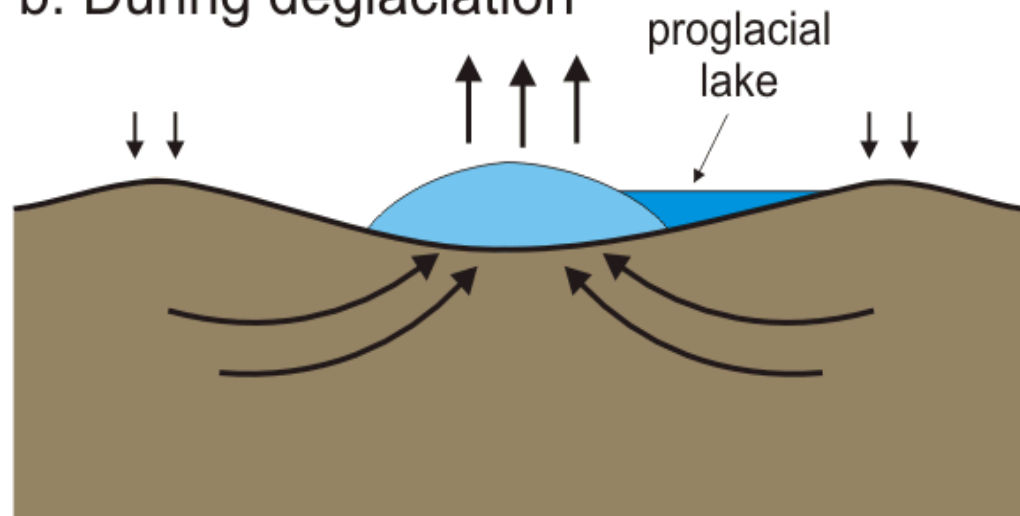
History of scientific development not covered
(see Whitehouse, 2018).

Glacial Isostasy

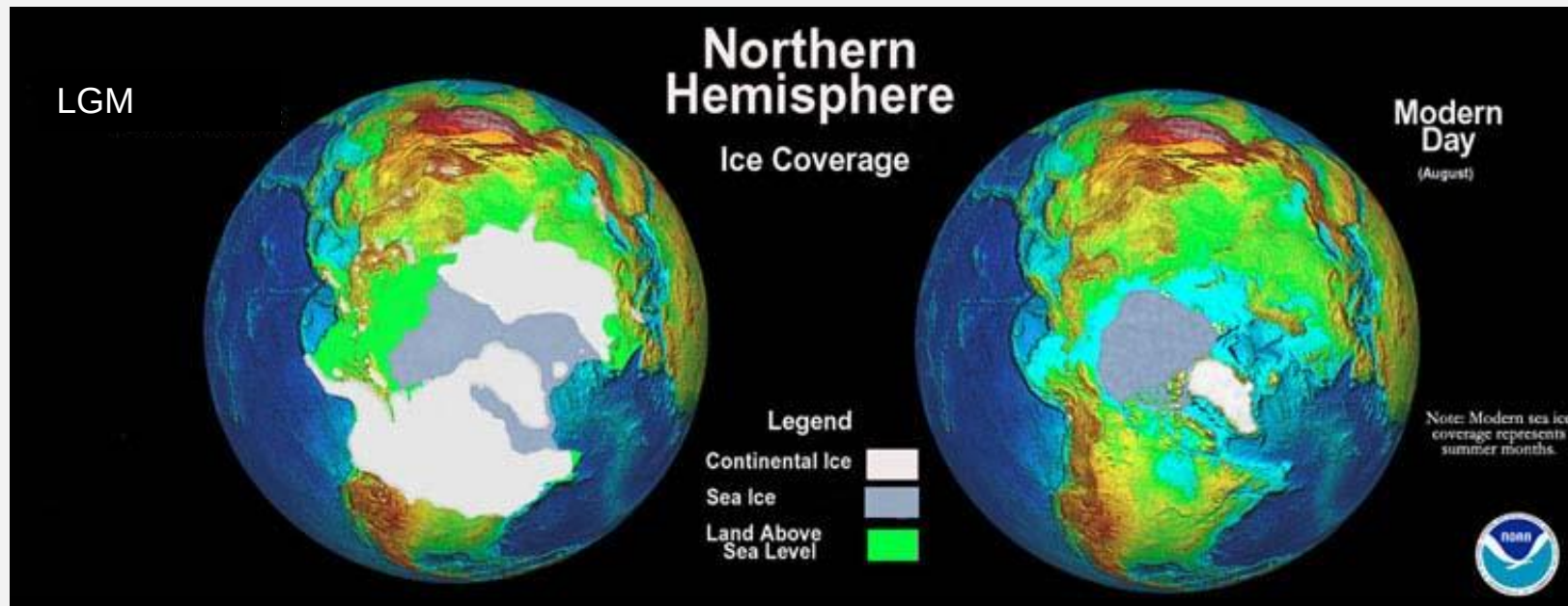
a. Peak glaciation



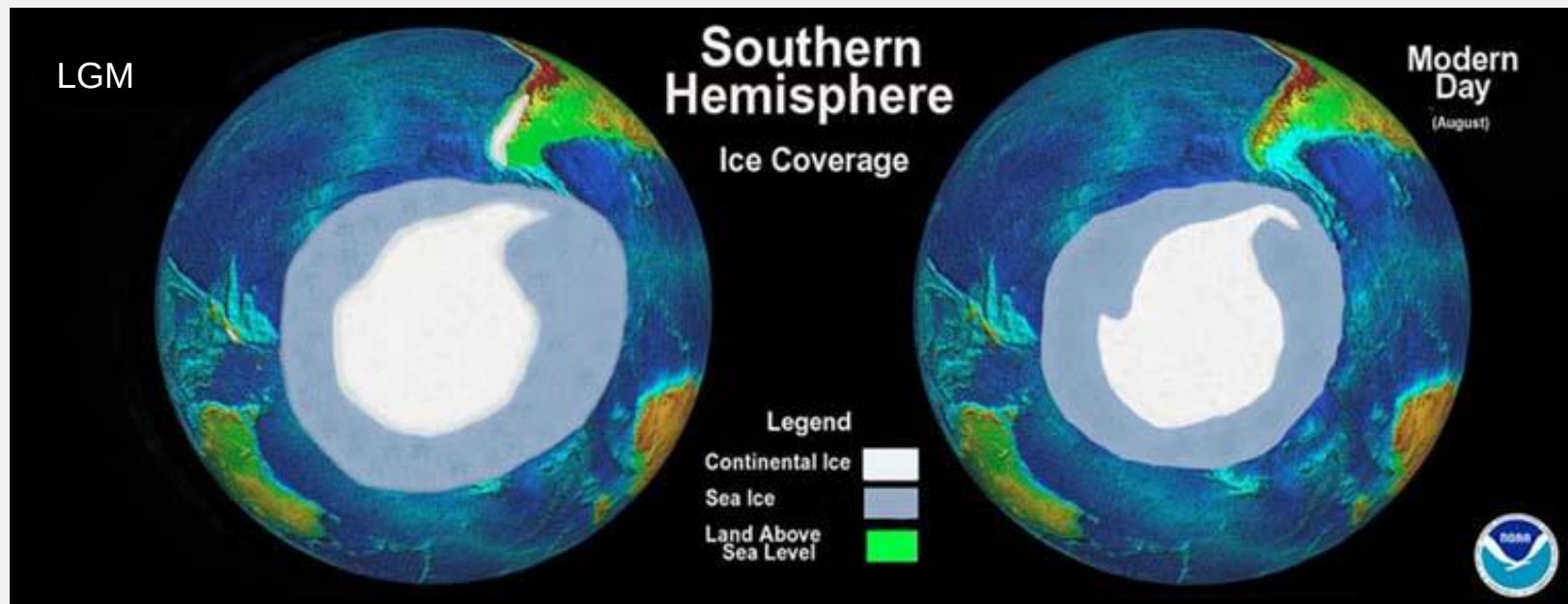
b. During deglaciation



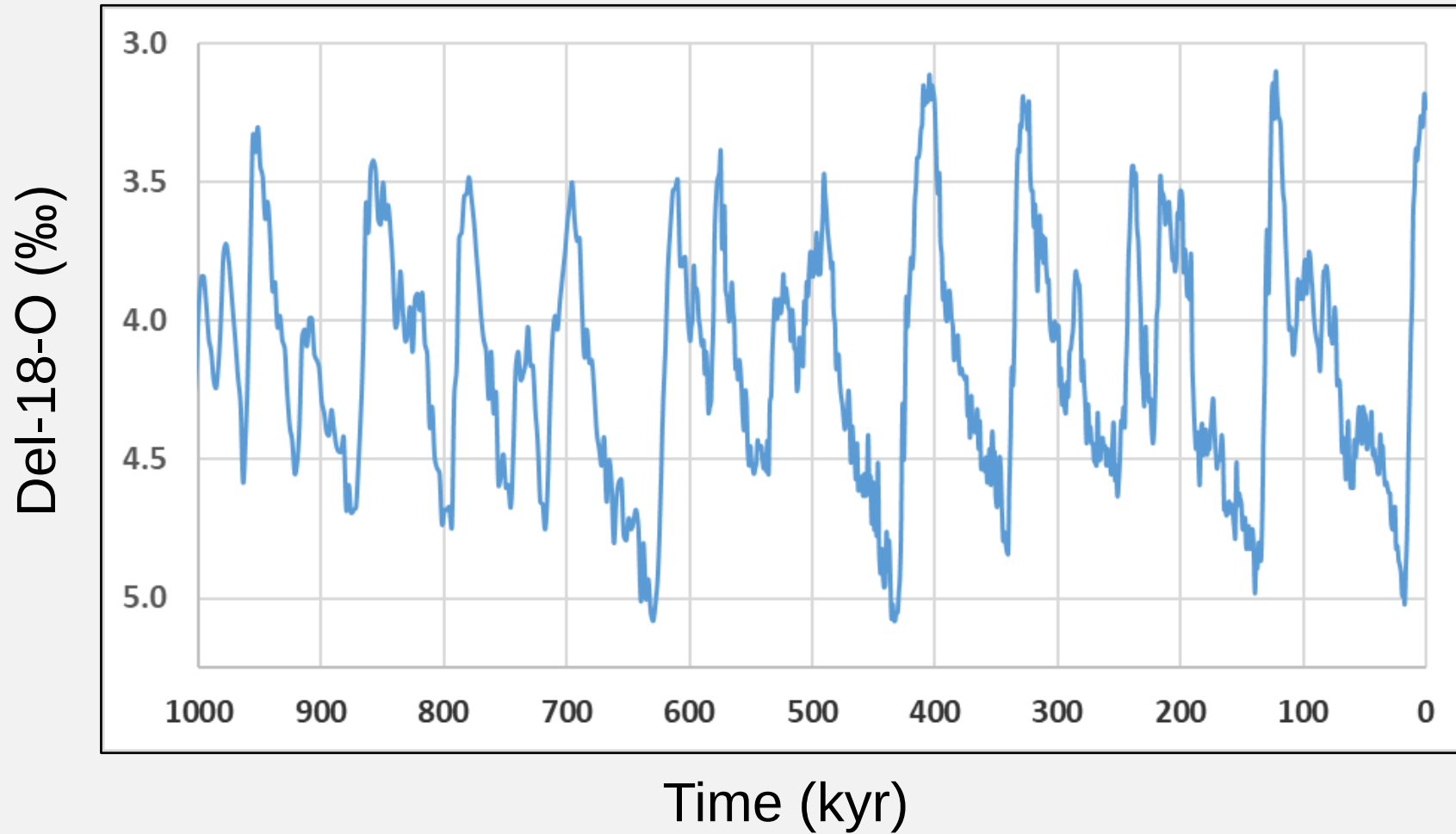
For a given loading history, the isostatic response is governed by the density and rheology of the Earth's interior.



Ice Extent at Last Glacial Maximum and Present-Day

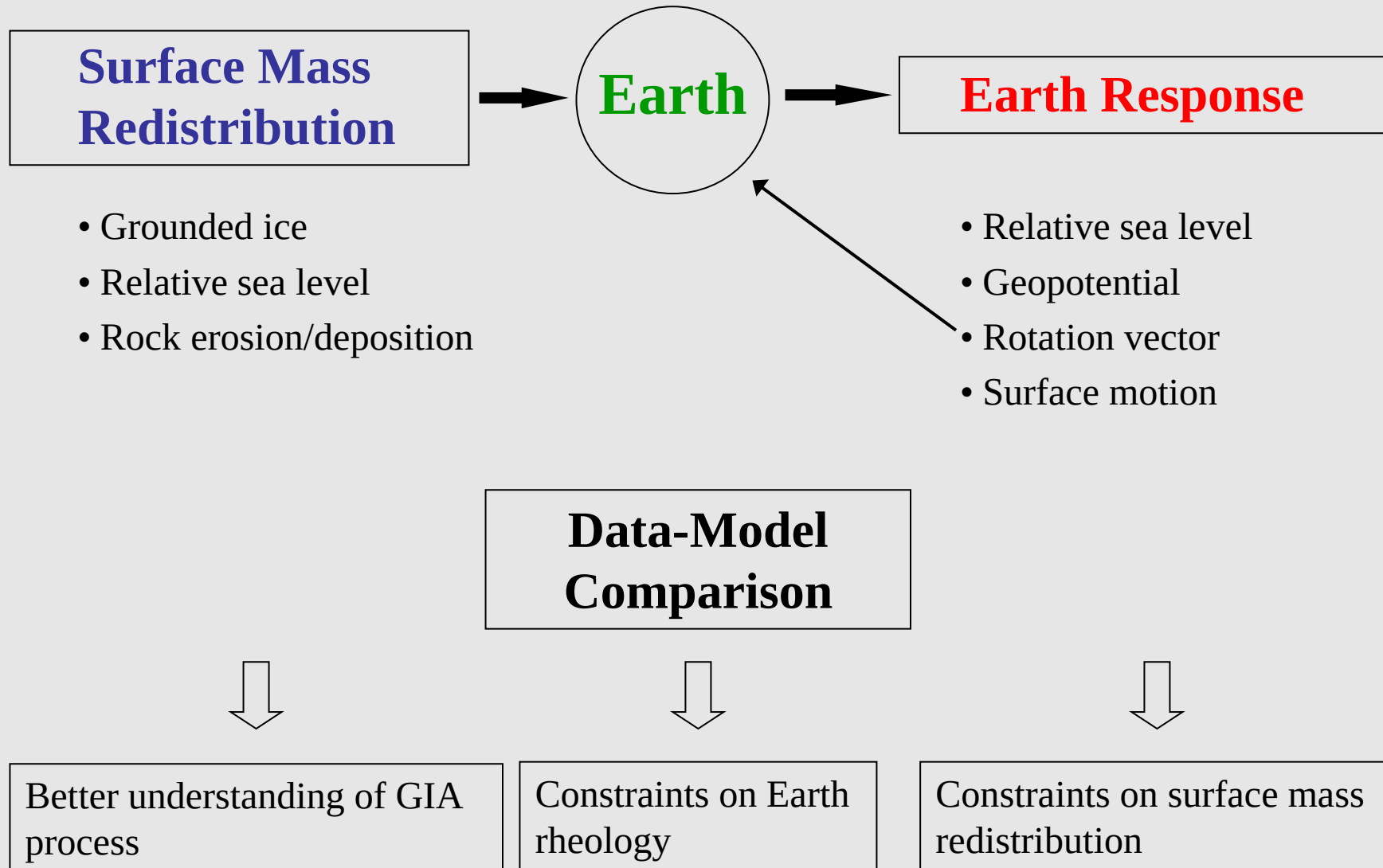


Time History of Ice: Marine Oxygen Isotope Data



Liseiki & Raymo (2005)

GLACIAL ISOSTATIC ADJUSTMENT



Some Observations Used in GIA Modelling

Paleo

Relative sea level
(shoreline deformation)

Lake levels (tilting)

Ice extent (geological
archives)

Geodetic

3D land motion (GPS)

Gravity changes (in-situ, GRACE)

Earth rotation changes (True Polar
Wander, Length of Day).

Relative sea level (tide gauges,
satellite altimetry)

Ice extent (satellite)

GIA Model Components

Earth Forcing

Surface Loading

Ice model

Sea-level model

**Erosion/Deposition
model**

Rotational Potential

Euler equations

Earth Model

Geometry

Spherical/Flat

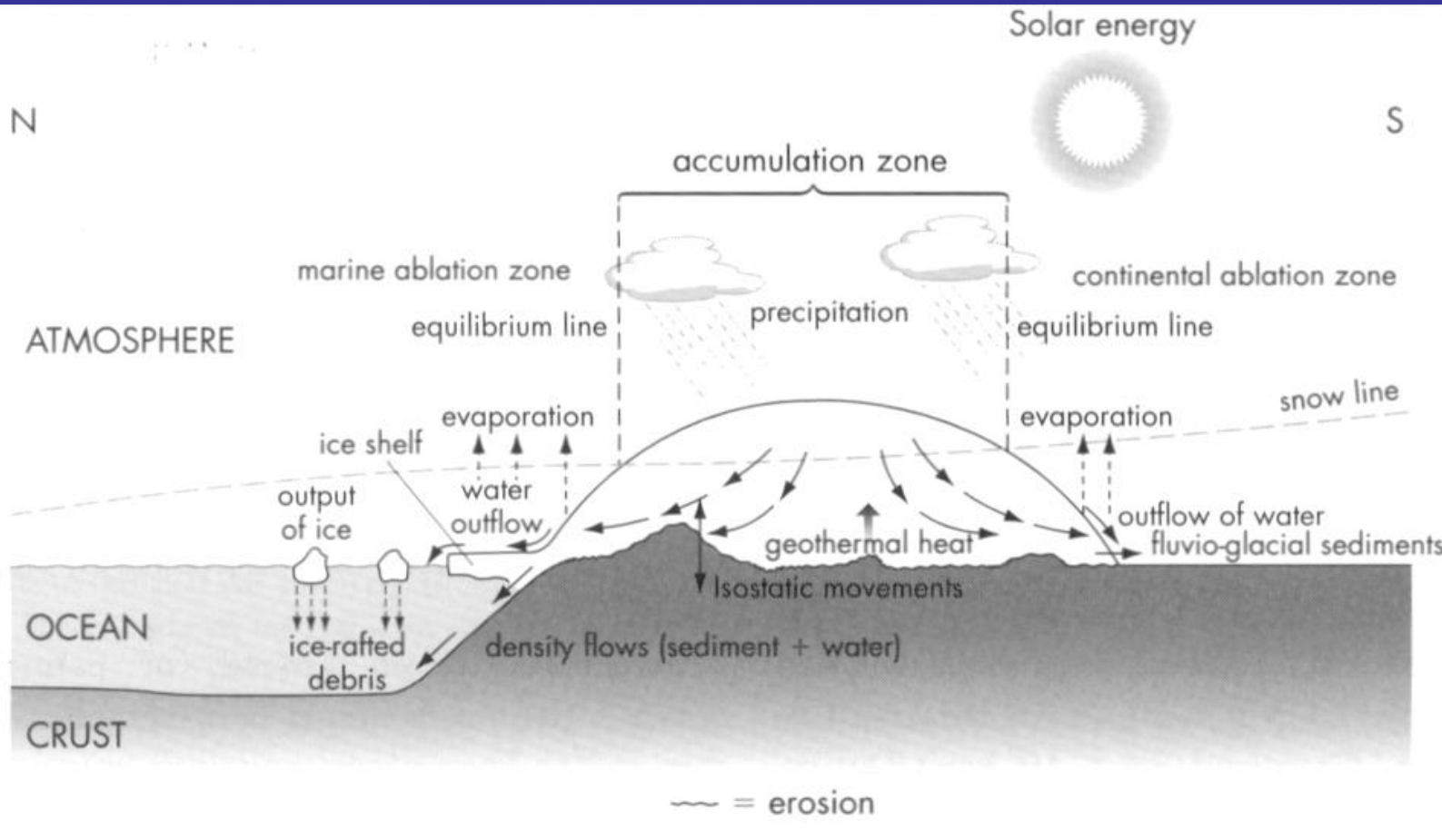
**Internal structure: 1D
& 3D**

Rheology

Viscoelastic

**Linear and non-
linear viscous
deformation**

Ice Loading History



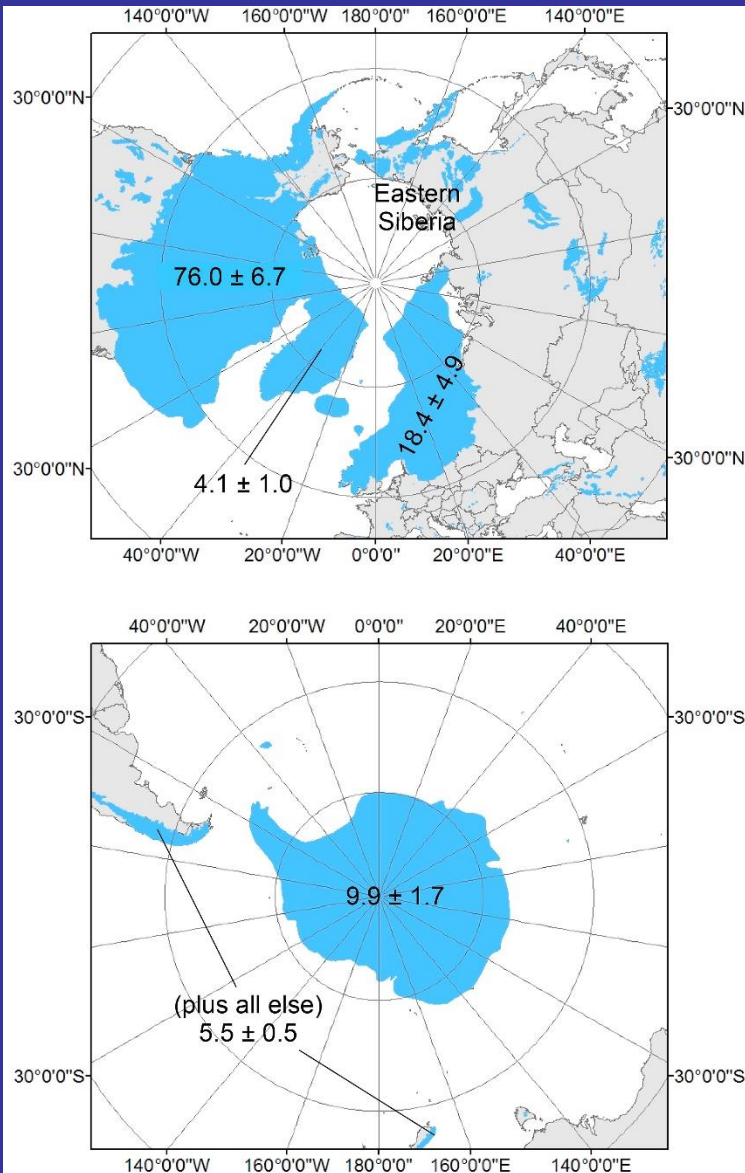
Direct data control from glacial geology and marine geophysics.

Indirect data control from GIA response (e.g., RSL, GPS).

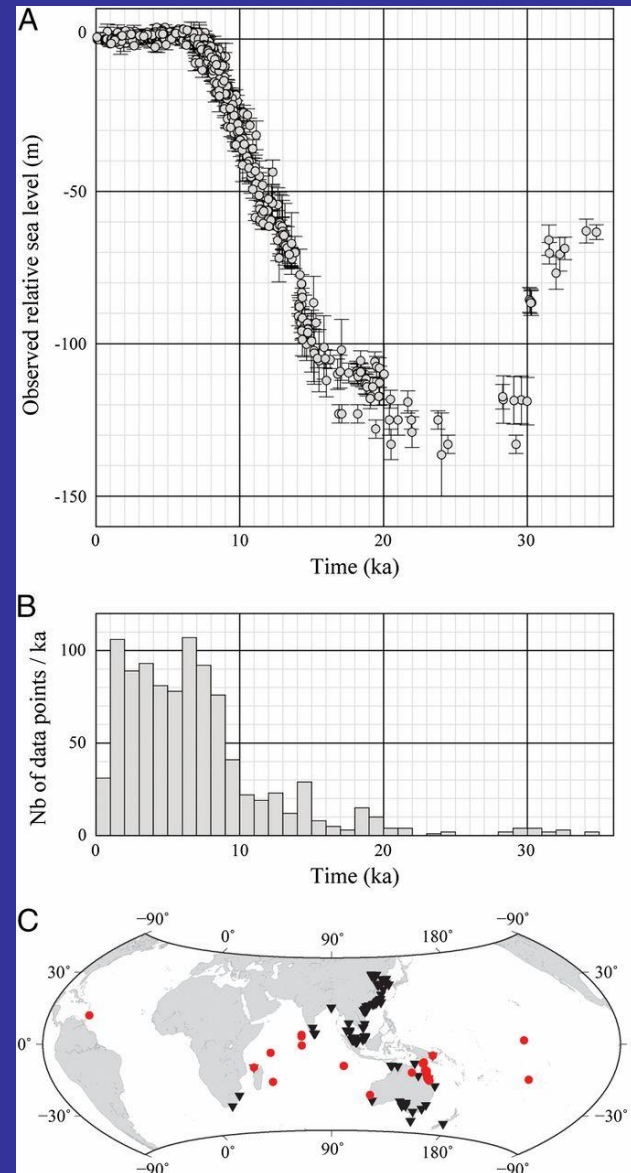
Constraints from glaciology (e.g., glaciological model simulations; simple ice physics).

For contemporary ice sheets can also use information from ice core records.

Global Ice Sheet Reconstruction



Simms et al. (2019)



Lambeck et al. (2014)

Determined by summing regional reconstructions.

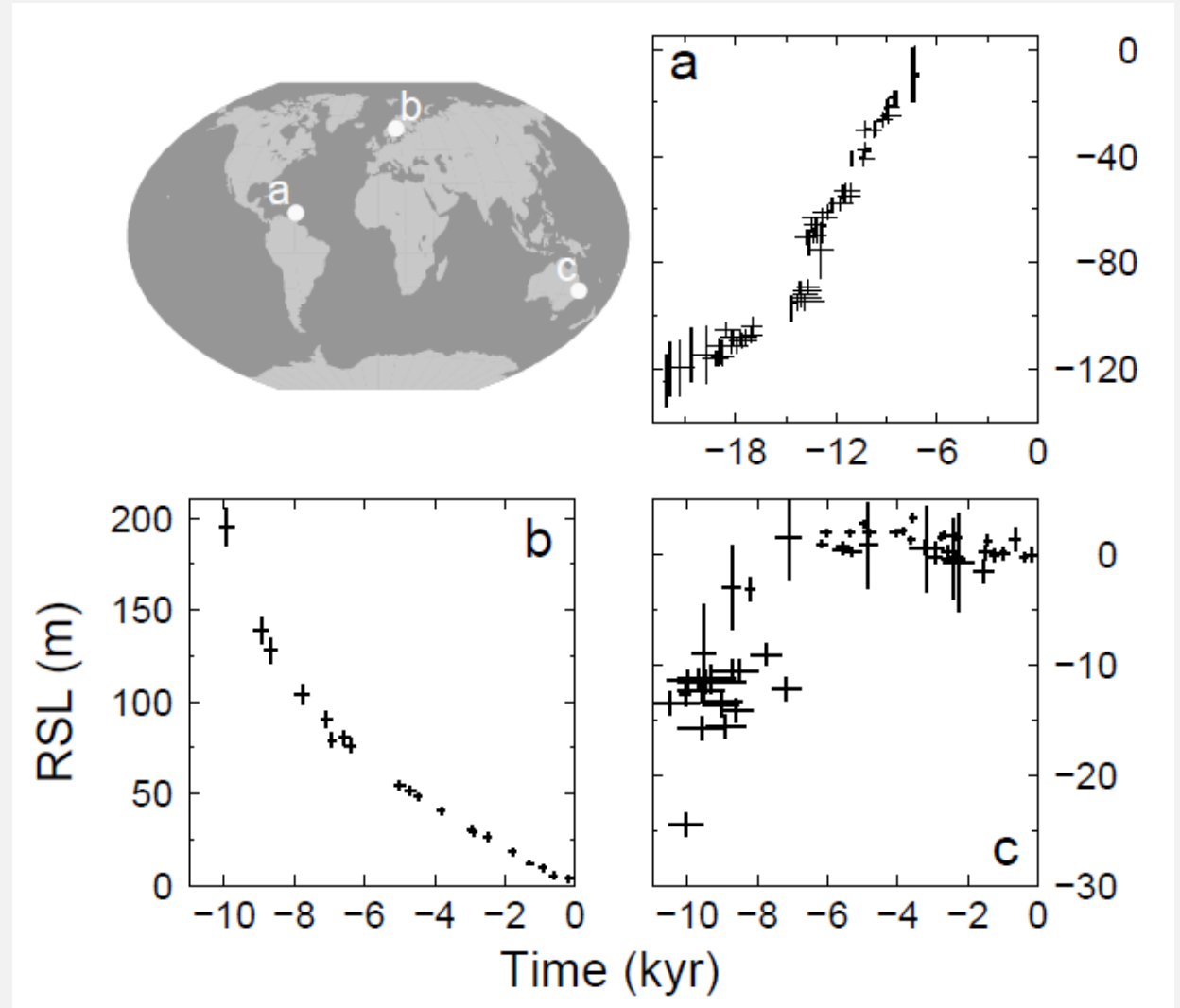
Volume of model reconstruction can be compared to that inferred from far-field RSL reconstructions.

At present these independent volume estimates do not compare well (see Clark and Tarasov, 2014; Simms et al., 2019).

Ocean Loading: Sea Level Change



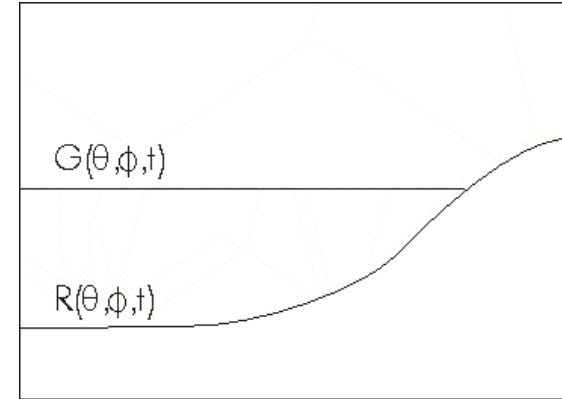
Courtesy of Steve Nerem



GIA-Related Departures From Barystatic Sea Level Change

- **Geoid perturbation,**

Gravity field perturbed **directly** by surface mass redistribution and changing rotational potential and **indirectly** by earth deformation caused by these forcings.



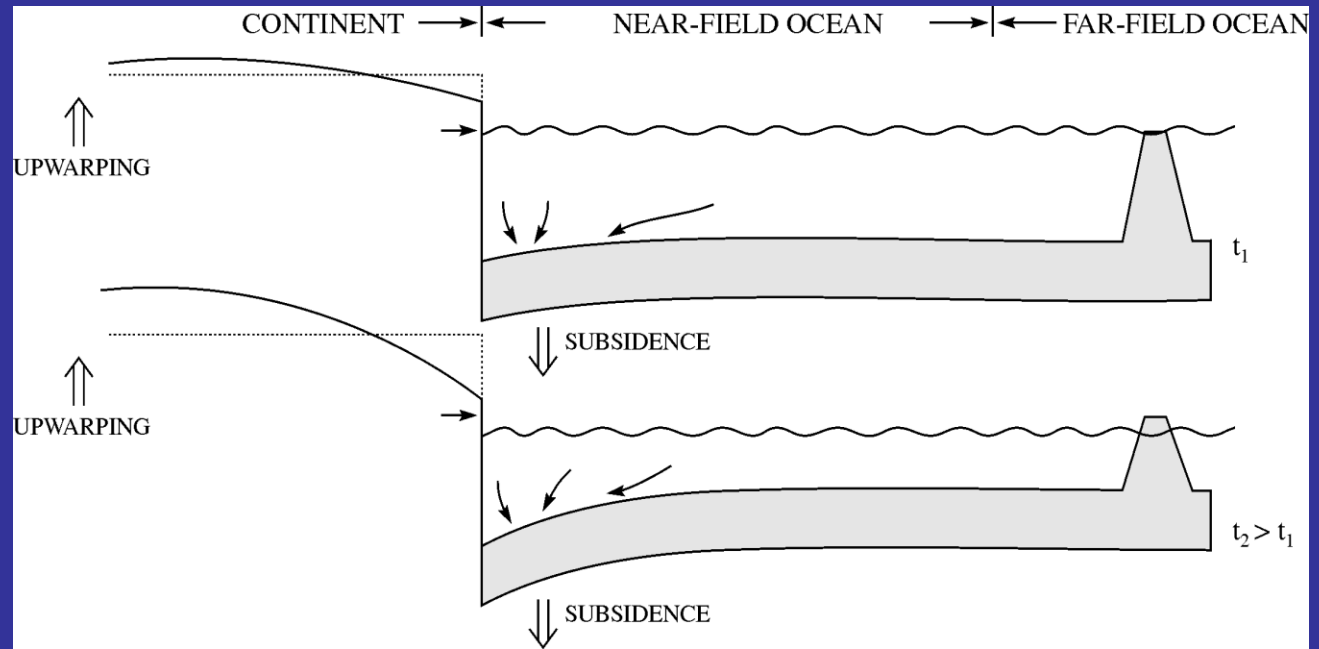
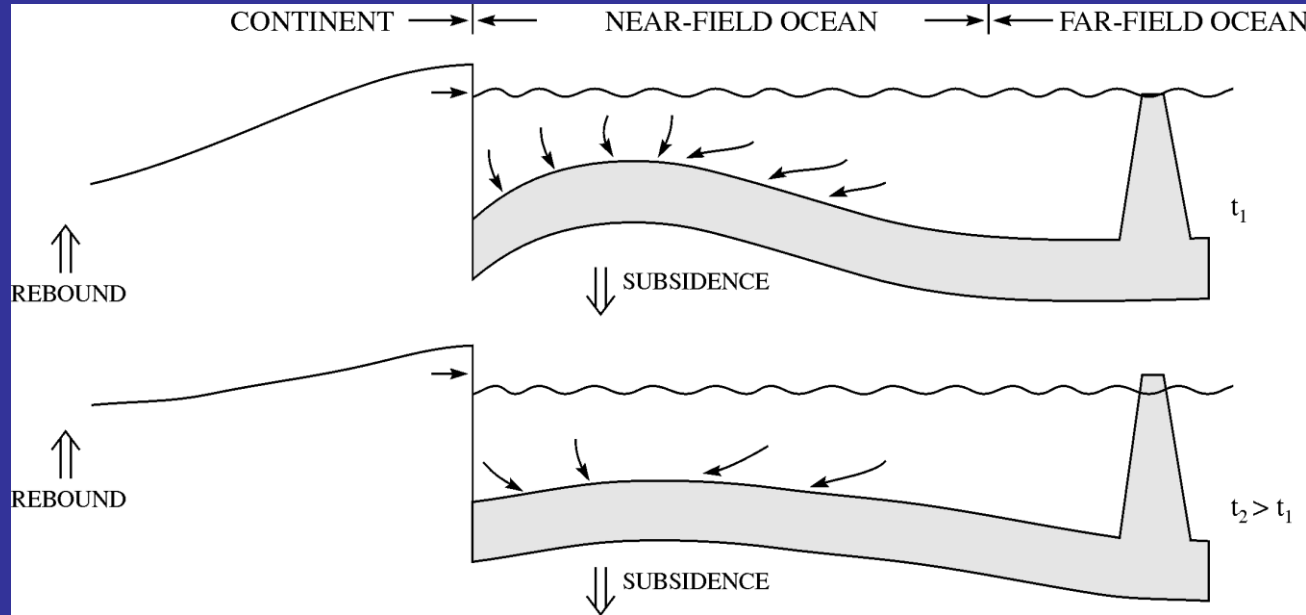
- **Solid surface perturbation,**

Vertical solid surface deformation associated with surface mass redistribution and changing rotational potential.

- **Ice-water mass conservation,**

Volumetric sea-level change must be consistent with mass lost/gained by ice sheets.

Syphoning

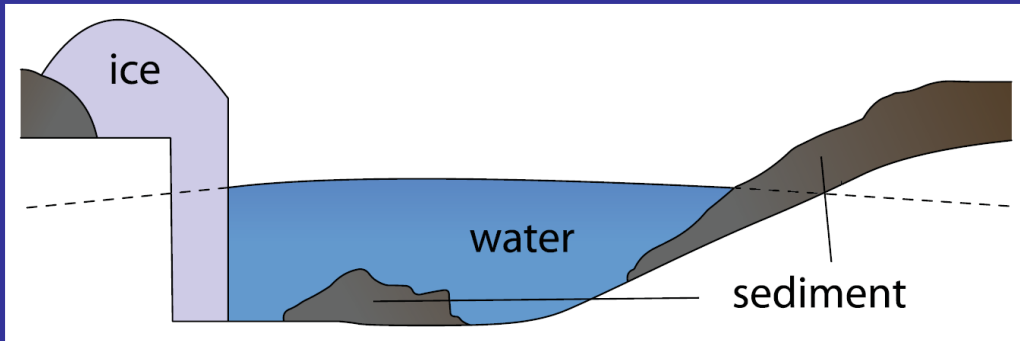


Theory of Sea-Level Change Associated With Ice-Ocean Mass Exchange

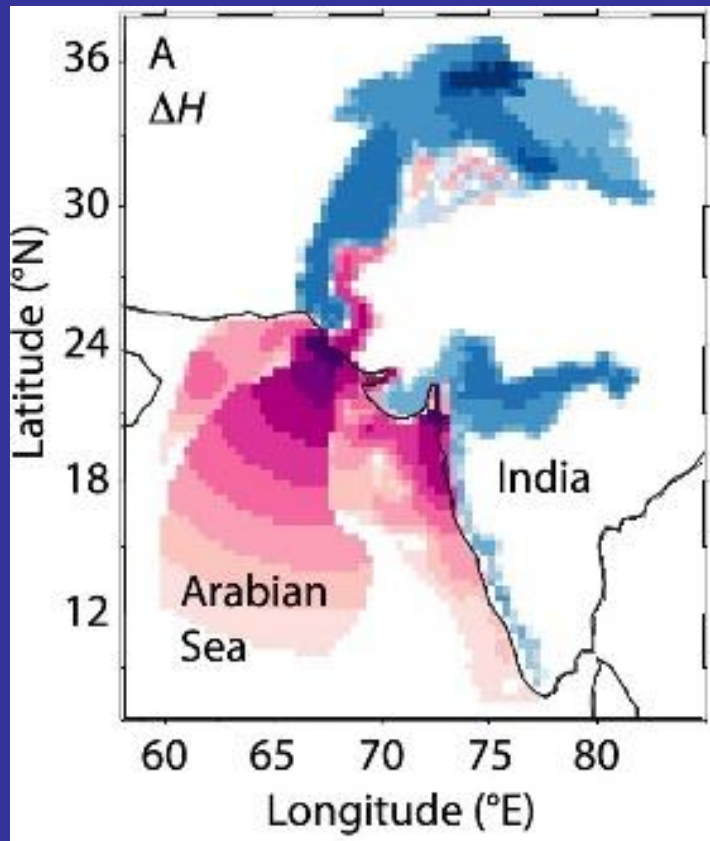
- Original paper by Farrell and Clark (1976) presented the sea-level equation which embodies the key theoretical elements.
- The sea-level equation considers the influence of ice-ocean loading and, more recently, perturbations in Earth rotation on sea level and time-variation in coastline position.
- For a review of the theoretical developments see Milne and Shennan (2013).

Erosion/Deposition Model

Dalca et al. (2013)



Sediment erosion/deposition influences sea-level and so including a sediment load requires that an extended sea-level equation be solved (Dalca et al, 2013).



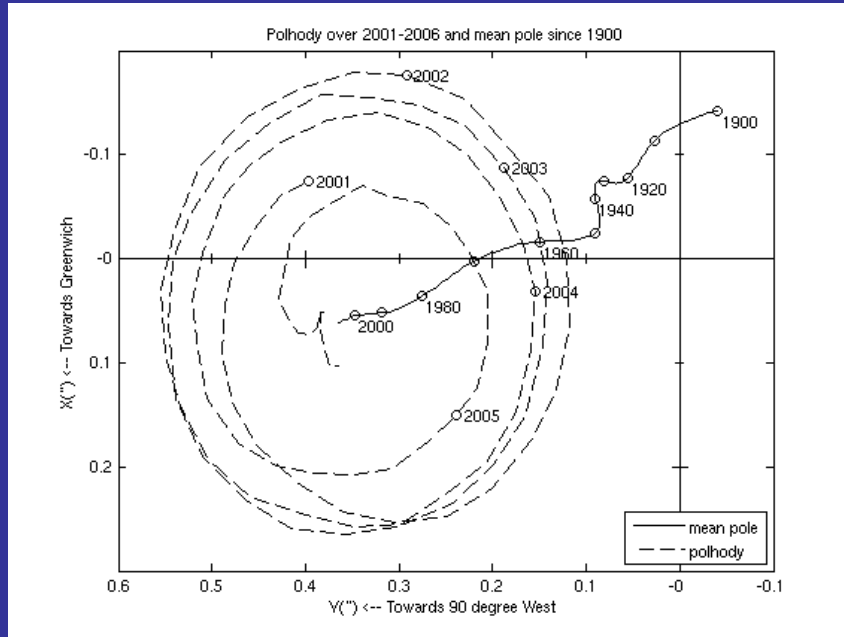
Influence of sediment loading most substantial near large river deltas (e.g., Arabian Sea, US Gulf coast – Ferrier et al., 2015; Kuchar et al. 2018).

Since this load is cumulative over time its impact is greater for older times (e.g., Pliocene – Moucha and Ruetenik, 2017).

Ferrier et al. (2015)

Changes in Earth Rotation

<https://www.iers.org/IERS/EN/Science/EarthRotation/PolarMotionPlot.html?nn=12932>

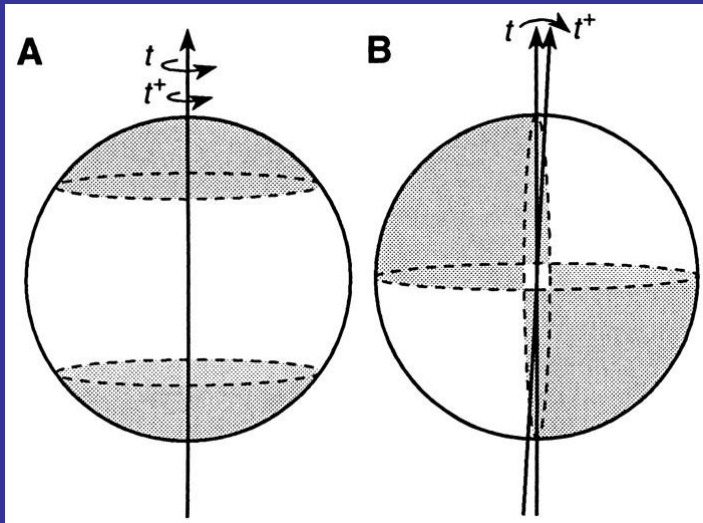


GIA changes the Earth's rotation vector (True Polar Wander and Length of Day).

Measurements of these changes can be used to constrain GIA models.

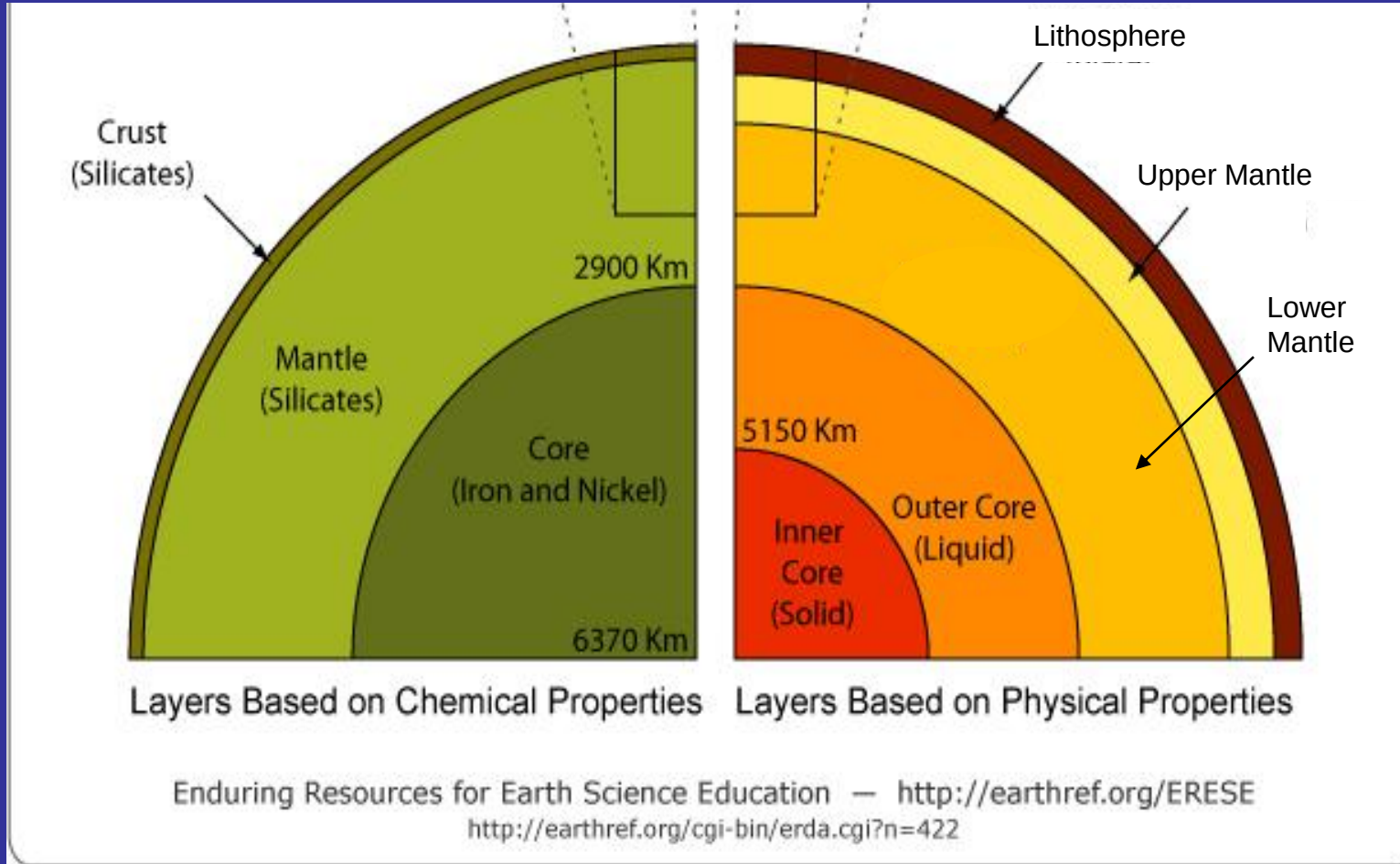
These changes also act as a secondary forcing via the 'pole tide' effect.

See Mitrovica and Wahr (2011).



Mitrovica et al. (2001a)

Earth Model (I-D)

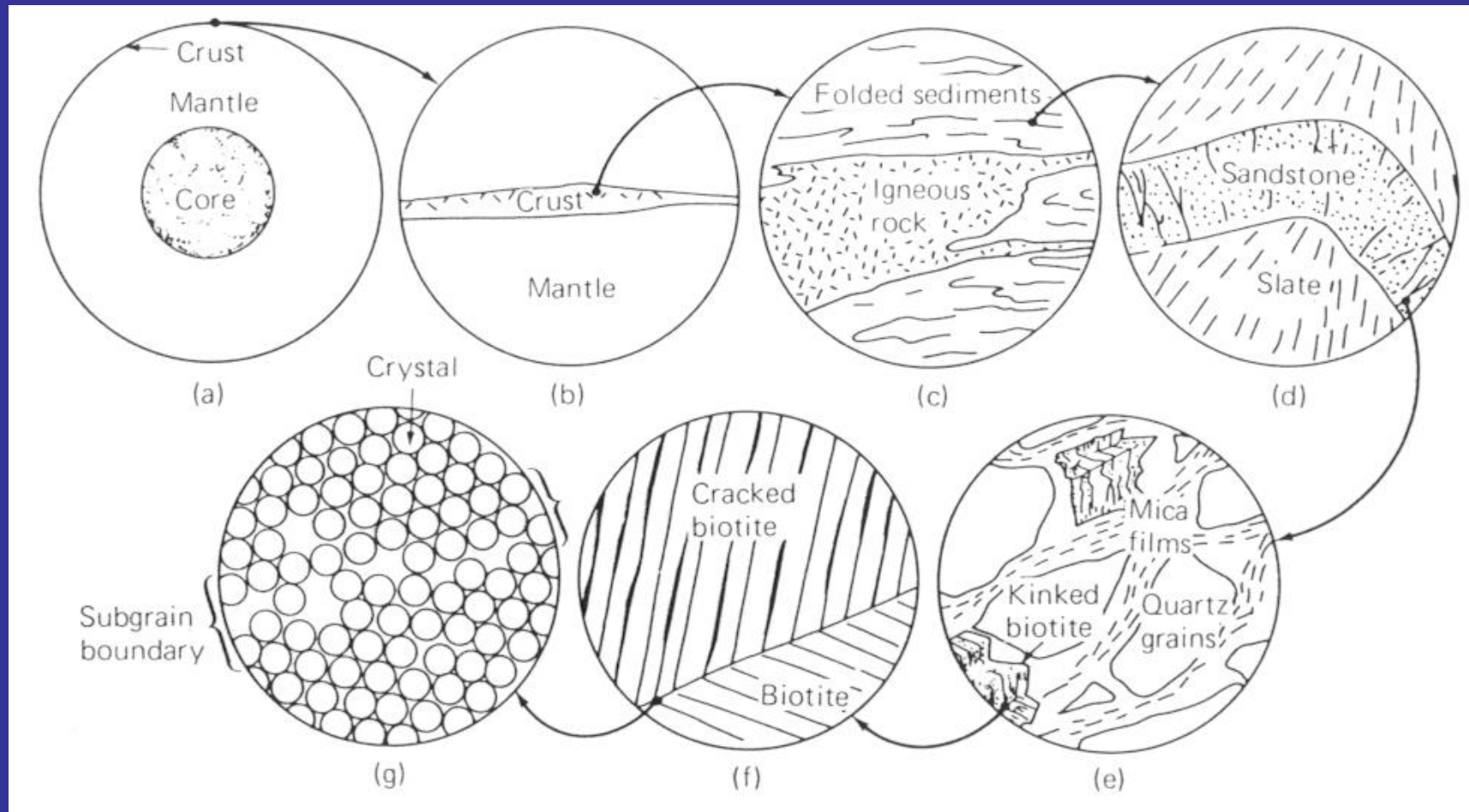


Density and elastic properties from seismic models.

Temperature and pressure are the dominant external controls on rock viscosity.

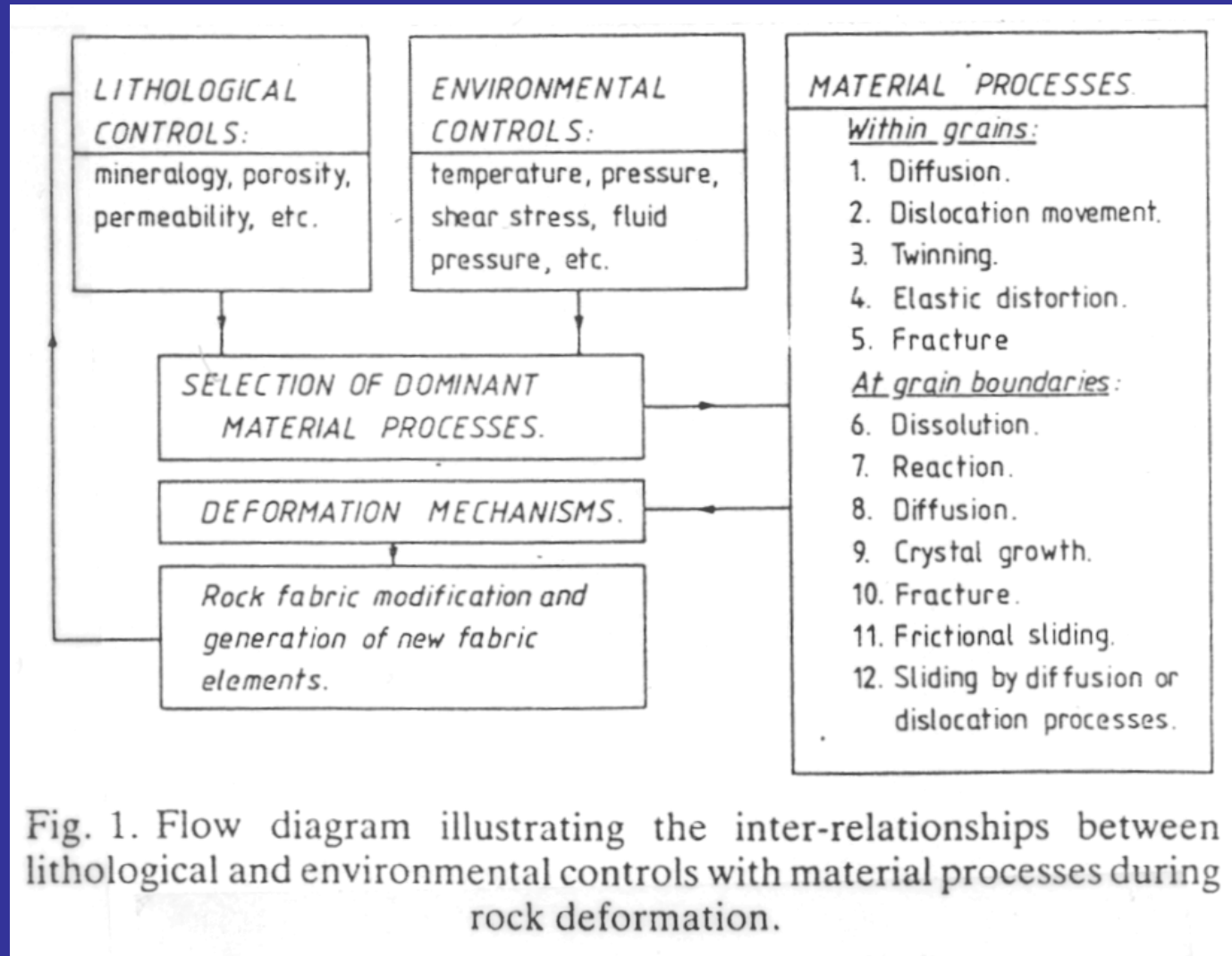
Viscosity structure can be estimated via results from lab deformation experiments and geophysical modelling.

Earth Model: Bulk rheological response governed by processes at grain and sub-grain scale

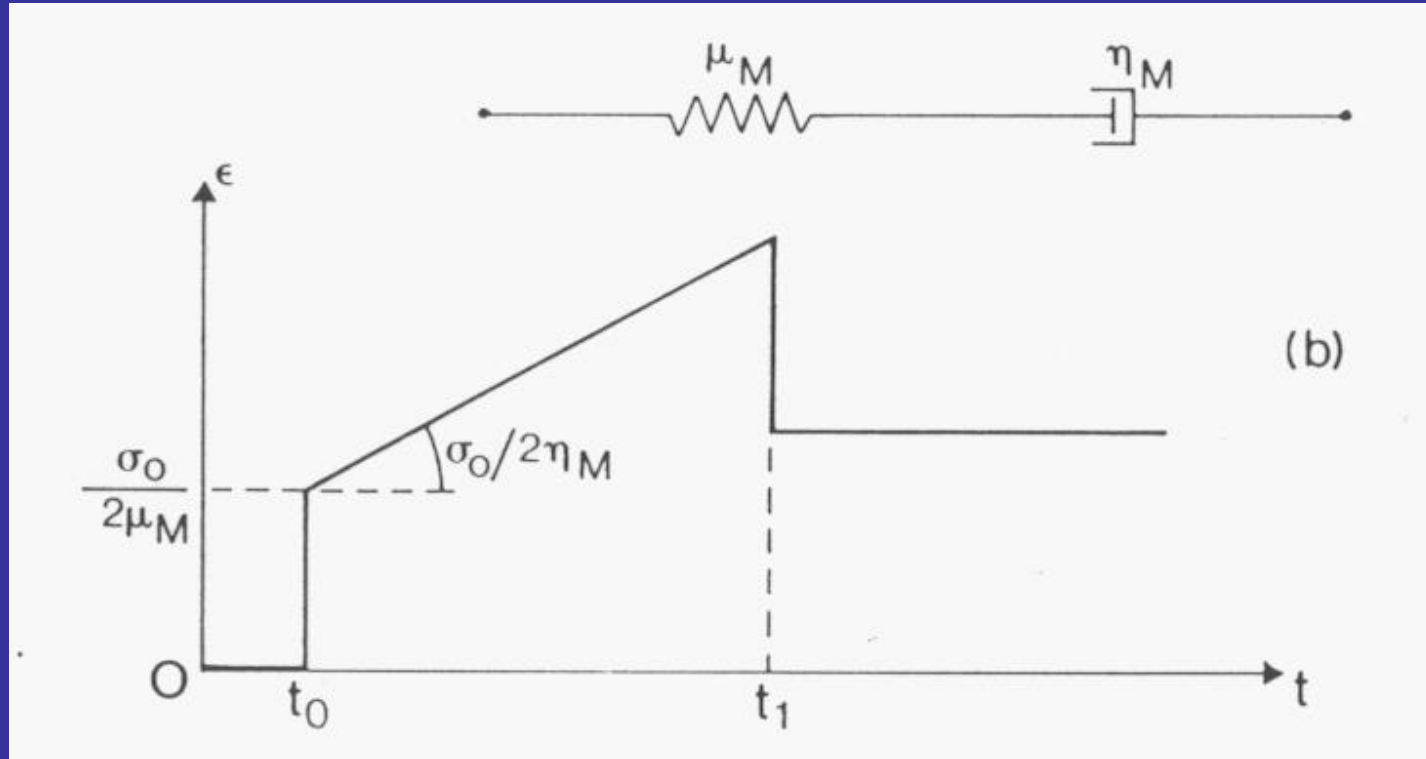


Means, *Stress and Strain*, Fig. 1.2

Processes and Parameters that Govern How Rocks Deform



Maxwell Model of Earth Rheology



Maxwell material is the 'default' rheological model for GIA modelling.

Ranalli (1995)

Common Models of Earth Rheology

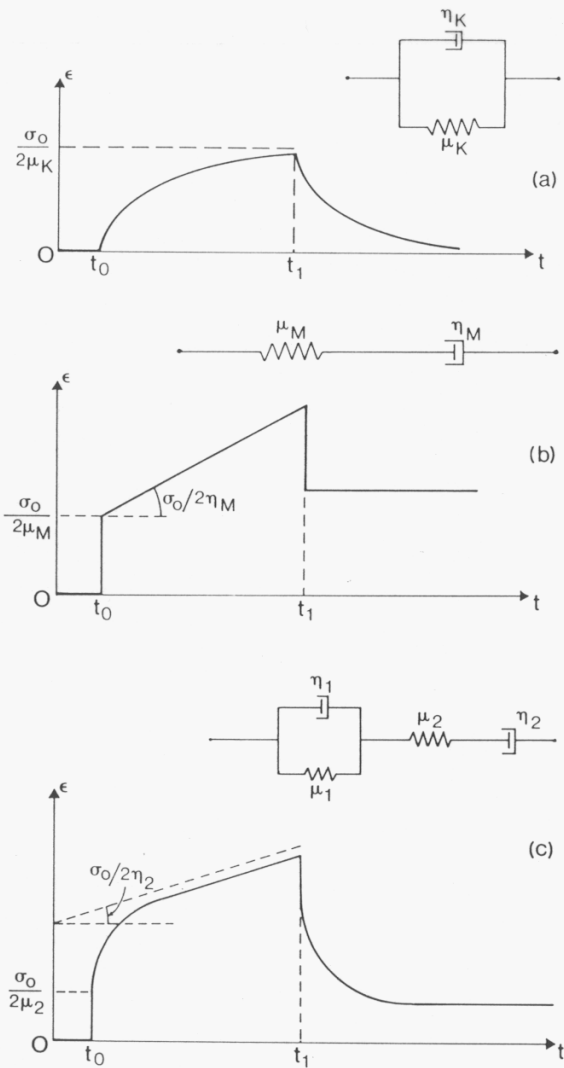


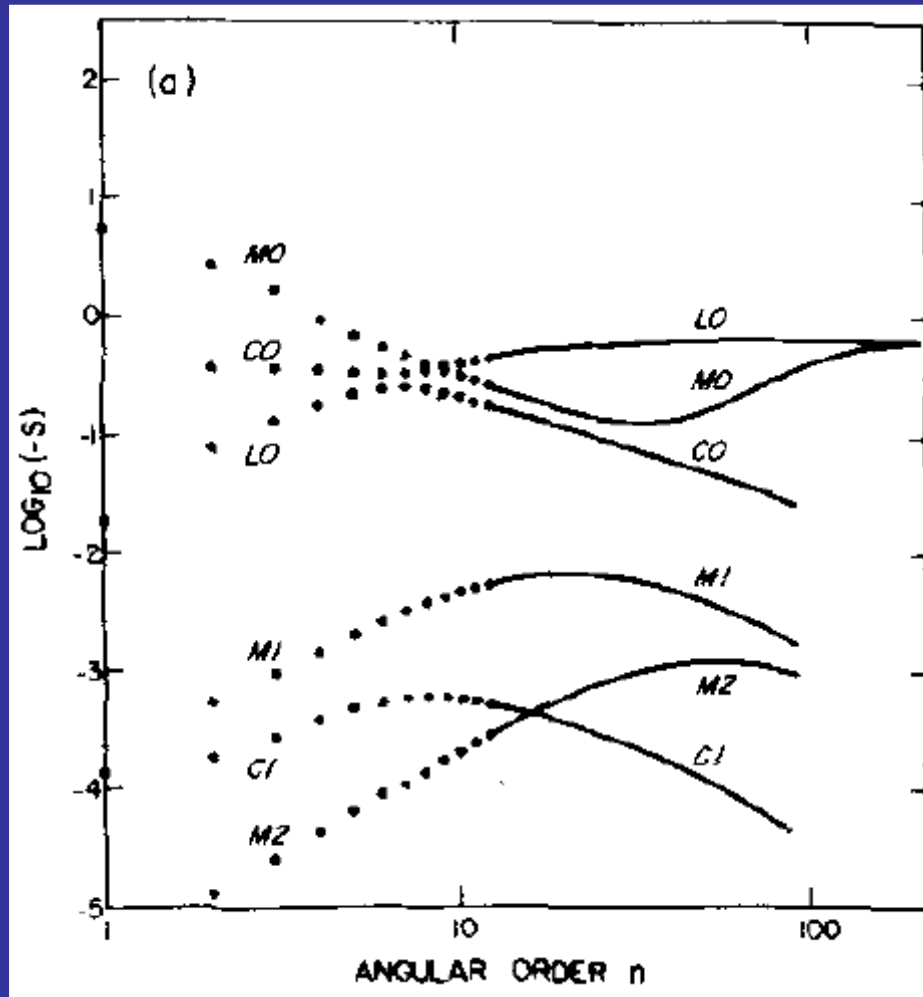
Figure 4.7 Rheological models and strain-time diagrams for (a) firmoviscous (Kelvin); (b) viscoelastic (Maxwell); and (c) general linear (Burgers) bodies.

Rock and mineral deformation experiments tell us Maxwell model is overly simplistic.

Earth materials exhibit a 'transient' behaviour. This behaviour can be captured by adding one or more Kelvin elements (Burgers model).

The non-elastic component of deformation can also often non-linearly related to the applied stress.

Normal Mode Formalism for Spherically Symmetric, Self-Gravitating, Visco-Elastic Earth Models



Consider the response of Earth model to an impulse forcing (Peltier, 1974).

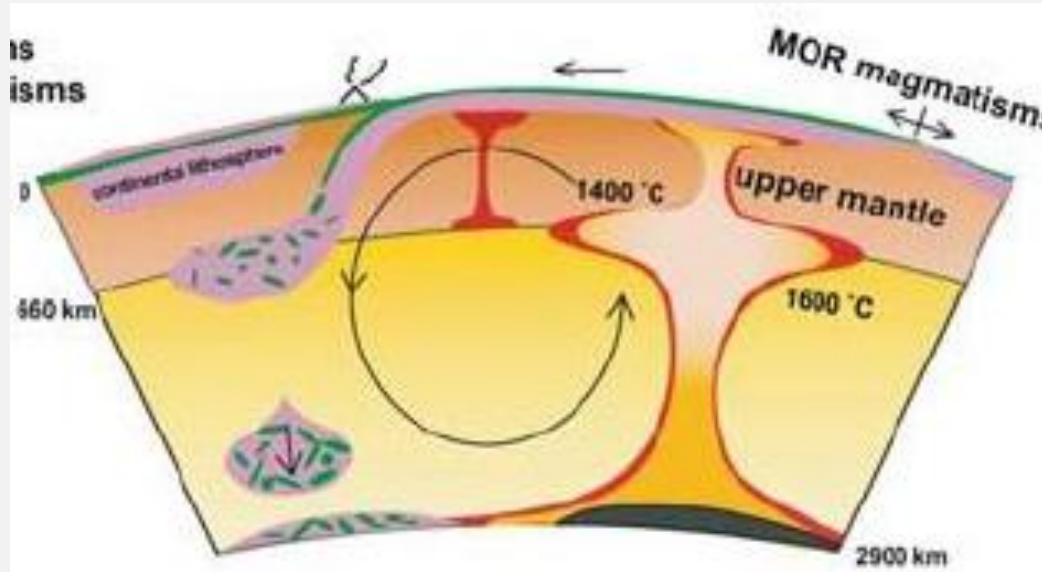
Simultaneously solve three differential equations: (i) momentum equation, (ii) continuity equation, (iii) Poisson's equation. Response of model to an impulse forcing.

Solution expressed in terms of Love numbers.

Peltier (1974), Wu & Peltier (1982)

Earth Model (3D)

Santosh et al. (2010)

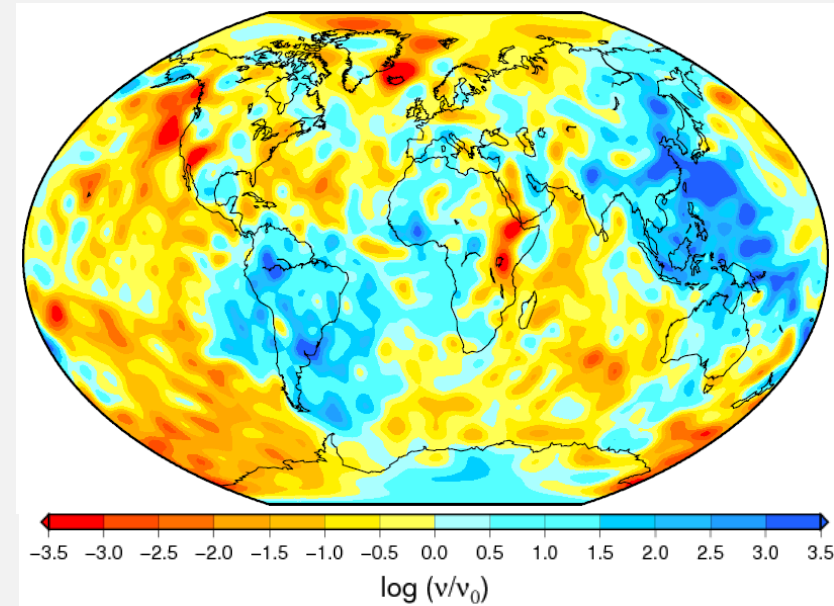
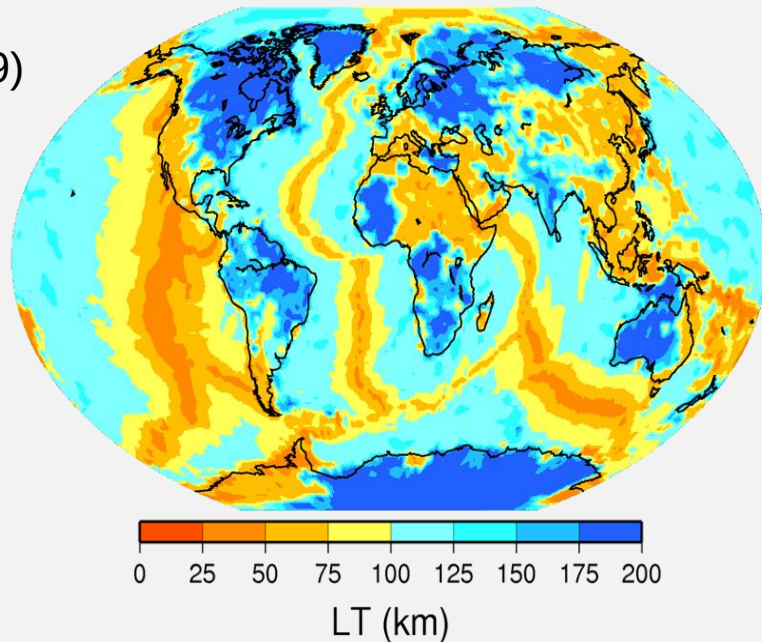


Lateral variations in Earth structure exist due to long-term Earth cooling.

Defining lateral structure relies on using information from other geophysical methods (e.g., seismology).

Differential equations (previous slide) solved numerically.

Afonso et al. (2019)



Based on
S40RTS
(Ritsema et
al. 2011)